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Lab 3 Report

I. Table with Estimated and Actual Resource Results

Resources	Estimate Used	Actual Used
Slice Registers	70	88
Register Latches	89	89
Clock Buffers	1	1
Number of IOs	48	50

II. Analyzing Resource Analysis

Estimating the number of IO ports can be done by counting the bits on the left hand side of a process. In the seven-segment controller, I counted six bits (the divided clock pulse, the reset signal, the clear signal, and three bits for the counter variable). In the top file, I counted 36 bits (32 for the array of logic vectors, the divided clock pulse, and the reset signal). Additionally, there are three bits within the process for the pulse generator (the clock signal, the reset signal, and the clear signal), and this module is implemented twice. This means the estimated number of IO ports should be 48. The actual number of IOB ports used is 50, meaning my estimate is close but not exact. I'm not sure where the other two bits would come from.

From the lab assignment, the number of flip-flops can be estimated by adding the number of bits that are affected by the rising clock edge of a process. In this lab, these values are the two instances of signal "pulseCnt" from the clock dividers (27 bits each), the 32 bits of the array for the character inputs, and 3 bits for the counter in the seven-segment controller. This leaves me with a total of 89 register latches, which matches the amount used. In my initial code, I found three latches, but these were removed when I realized that I forgot to include the "rising edge" keyword in the process.

My estimate for the number of clock buffers is one, as there is one clock that is being divided in this assignment. This number easily matches the amount of clock buffers that were actually used, which was one. Slice registers, however, were the element I had the most trouble estimating. I thought it had to do with the number of bits being assigned outside of the process. In the top module, I counted 32 for the array that determined the seven-segment outputs, 4 for the assignment of "binary_sw," which made the first four switches their own logic value, 16 for the designation of LEDs to switches, and 2 for routing the clock and reset values, which led to a total of 70. However, this is under the actual value of 88 slice registers. I believe that I need help to find a better understanding of this estimation process.